

REAL AND COMPLEX ANALYSIS (3-1-0-4)

COURSE CODE-MC222

Course placement: This is a core course for second year students of BTech MnC program.

Course format: It is 3 hours lecture every week and 1 tutorial per week.

Course content: The primary aim of this course is to introduce students to the fundamentals of Real and complex analysis. In this course we cover the following chapter:

Basic Notions: Countable and uncountable sets, Definition and Examples of Metric spaces, Open Balls and Open Sets. (**Three lectures**).

Convergence: Convergent Sequences, Limit and Cluster Points, Cauchy Sequences and Completeness, Bounded Sets, Dense Sets, Basis, Boundary of a Set, Series and convergence criteria for series of positive real numbers. (**Five lectures**).

Continuity: Continuous Functions, Equivalent Definitions of Continuity, Topological Property, Uniform Continuity, Limit of a Function, Open and closed maps. (**Four lectures**).

Compactness: Compact Spaces and their Properties, Continuous Functions on Compact Spaces, Characterization of Compact Metric Spaces. (**Three lectures**).

Connectedness: Connected Spaces, Path Connected spaces. (**Two lectures**).

Complete Metric Spaces: Examples of Complete Metric Spaces, Baire Category Theorem. (**Two lectures**).

Complex Numbers and the Complex Plane: Complex Numbers and their Properties Complex Plane, Polar Form of Complex Numbers, Powers and Roots, Sets of Points in the Complex Plane. (**Two lectures**).

Complex Functions and Mappings: Complex Functions, Complex Functions as Mappings, Linear Mappings, Special Power Functions, The Power Function, Reciprocal Function, Limits and Continuity. (**Two lectures**).

Analytic Functions: Differentiability and Analyticity, Cauchy-Riemann Equations, Harmonic Functions, Applications. (**Three lectures**).

Integration in the Complex Plane: Real Integrals, Complex Integrals, Cauchy-Goursat theorem, Independence of Path, Cauchy's Integral Formulas and Their Consequences, Cauchy's Two Integral Formulas, Some Consequences of the Integral Formulas. Applications. (**Five lectures**).

Series and Residues: Sequences and Series, Taylor Series, Laurent Series, Zeros and Poles, Residues and Residue Theorem, Some Consequences of the Residue Theorem. (**Five lectures**).

Conformal Mappings: Conformal Mapping, Linear Fractional Transformations. (**Two lectures**).

Suggested Textbook/references

- Topology of metric spaces, S. Kumaresan.
- A basic course in real analysis, S. Kumaresan and A. Kumar.
- A first course in complex analysis with applications, D. G. Zill and P. D. Shanahan.
- Complex analysis, T. Gamelin.
- Complex Analysis, L. V. Ahlfors.

Assessment method: Two in-semester examinations(30% each), End-semester examination(40%).

Course outcome: After successful completion of the course, students will be able to:

- Understand the ideas of open and closed sets, convergence, continuity, completeness and compactness in the topological spaces.
- Understand a new perspective on many of the ideas studied in Real Analysis.
- Represent complex numbers algebraically and geometrically
- Analyze limit, continuity and differentiation of functions of complex variables.
- Understand Cauchy-Riemann equations, analytic functions and various properties of analytic functions.
- Understand Cauchy theorem and Cauchy integral formulas and apply these to evaluate complex contour integrals.
- Represent functions as Taylor and Laurent series; classify singularities and poles; find residues and evaluate complex integrals using the residue theorem.
- Understand conformal mapping.
- Apply the knowledge of mathematics, science, engineering fundamentals to the solution of complex engineering problems (P1).
- Understand and create mathematical arguments for solving problems (P2).
- Develop skills towards mathematical modeling and analysis of engineering problems (P3, P4).

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
x	x	x	x								